

January 2026 CSE 406: Cyber Security Sessional

Online 1: Buffer Overflow

Section: C2

Time: 25 minutes

Objective

You are given a vulnerable C program (`target.c`) that allocates data on the **heap** using `malloc()`. Your task is to overflow a heap-allocated buffer to corrupt an adjacent heap-allocated **function pointer**, redirecting execution to a secret function that spawns a shell.

Environment Setup

```
seed@ubuntu:~$  
sudo sysctl -w kernel.randomize_va_space=0  
sudo ln -sf /bin/zsh /bin/sh
```

Student ID Configuration

Open `target.c` and replace `<LAST_3_DIGITS_OF_YOUR_ID>` in the `#define STUDENT_ID` line with the **last 3 digits** of your Student ID.

Compilation

```
seed@ubuntu:~$  
gcc -m32 -o target -fno-stack-protector target.c  
sudo chown root target  
sudo chmod 4755 target  
touch badfile  
gcc -m32 -o target_dbg -g -fno-stack-protector target.c
```

Task

Write a Python exploit script (`exploit.py`) that generates a `badfile` such that when the program is run, it prints:

```
seed@ubuntu:~$  
HACKED! Secret action triggered!
```

And spawns a root shell.

Restrictions

1. **Do NOT modify `target.c`** (except for setting your `STUDENT_ID`)
2. You must corrupt the heap-allocated function pointer to redirect execution to the secret function — direct calls are not sufficient
3. Your exploit must work on the `setuid` binary (`target`), producing a root shell

Submission

1. Create a folder with your student ID as name (i.e., 2105185)
2. Put `target.c` and `exploit.py` inside it.
3. Zip the folder and upload it.